This section will discuss the importance of open source frameworks and two critical advantages they provide – transparency and flexibility.

Open source features

How open source features relate to the Data Intelligence Platform is unique. This uniqueness lies in the concepts of openness and transparency, often referred to as the "glass box" approach by Databricks. It means that when you use the platform to create assets, there's no inscrutable black box that forces you to depend on a specific vendor for usage, understanding, or storage. A genuinely open lakehouse architecture uses open data file formats to make accessing, sharing, and removing your data simple. Databricks has optimized the managed version of Apache Spark to leverage the open data format Delta (which we'll cover in more detail shortly). This is one of the reasons why the Delta format is ideal for most use cases. However, nothing stops you from using something such as the CSV or Parquet format. Furthermore, Databricks introduced **Delta Lake Universal Format** (**Delta Lake UniForm**) to easily integrate with other file formats such as Iceberg or Hudi. For more details, check out the *Further reading* section at the end of this chapter.

Figure 1.2 illustrates the coming together of data formats with UniForm.

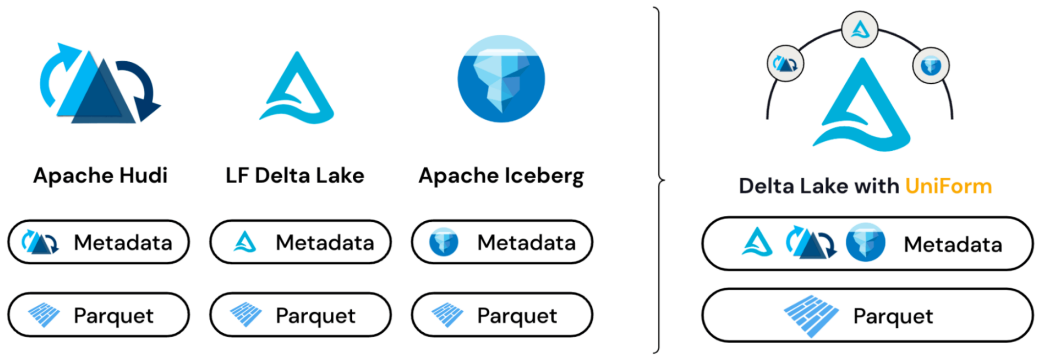


Figure 1.2 – Delta Lake UniForm makes consuming Hudi and Iceberg file formats as easy as consuming Delta

The ability to use third-party and open source software fuels rapid innovation. New advances in data processing and ML can be quickly tested and integrated into your workflow. In contrast, proprietary systems often have longer wait times for vendors to incorporate updates. Waiting for a vendor to capitalize on open source innovation may seem rare, but it is the rule rather than the exception. This is especially true for data science. The speed of software and algorithmic advances is incredible. Evidence of this frantic pace of innovation can be seen daily on the Hugging Face community website. Developers share libraries and models on Hugging Face; hundreds of libraries are updated daily on the site alone.